

KEY OPERATION

OPERATION
NEPTUNE SPEAR
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The killing of the al-Qaeda (see p.408) leader Osama bin Laden by US SEAL Special Forces, on 2 May 2011, was achieved through a combination of intelligence and technology. Bin Laden's Pakistani hideout was observed by satellites, and then by the covert Sentinel stealth UAV. The US SEALs were transported by stealth UH-60 Black Hawk helicopters and carried the latest equipment, including suppressed carbines, night-vision goggles, and body armour.



▲ Osama bin Laden was the founder of al-Qaeda, the jihadist group responsible for the terrorist attacks on the US, on 11 September 2001 (see p.408).

▼ XM25 GRENADE LAUNCHER

A US Army soldier tests an XM25 grenade launcher in Afghanistan. The XM25 uses computerized optical sights, and fires 25mm grenades set to explode in mid-air at or near the target, and also pre- or post-impact.



KEY DEVELOPMENT

THE HIGH-TECH
BATTLEFIELD

The pursuit of technology to improve battlefield performance continues at an ever-increasing pace. In a race currently led by the US, all the major military nations are incorporating the latest scientific and technological advances into their weapon systems.

The development of the unmanned aerial vehicle (UAV) has brought about a profound change to the way battles are fought. Towards the end of the 20th century, the UAV had already established itself as a valuable reconnaissance tool; in particular, the RQ-4 Global Hawk and the Predator and Reaper drones have revolutionized the process of gathering strategic and tactical intelligence. Post-2000, the UAV has also assumed the role of hunter-killer. The MQ-9 Reaper, for example, can be armed with up to 14 Hellfire air-to-ground missiles or two Paveway II laser-guided bombs. It has seen extensive use in Afghanistan, where it is controlled via a real-time satellite link from an air force base in Nevada, some 9,600km (6,000 miles) away. A further advantage of the UAV is its airborne endurance: the Israeli Eitan, for example, can stay airborne for up to 70 hours.

UNMANNED VEHICLES ON LAND

As part as the American-led move towards the “automated battlefield”, unmanned vehicles are also used on land. This began with remote-controlled bomb-disposal machines, but in recent years, larger, more complex vehicles have been produced, including the remote-controlled Black Knight tank, which is based on the M2 Bradley infantry combat vehicle. An intriguing development has been the Israeli remote-controlled robotic snake. Used for surveillance in confined, hostile environments, it is equipped with a camera and microphone.

Progress made in applying stealth technology to standard front-line aircraft (such as the US F-35) has now been extended to warships and armoured vehicles.

Developments in Adaptive Camouflage allow such high-profile ground objects as a main

battle tank to be hidden from infra-red sensors. And, heading into the realms of science fiction, research has also been conducted into a material called Metaflex, which may be able to hide soldiers behind a kind of “invisibility cloak”.

